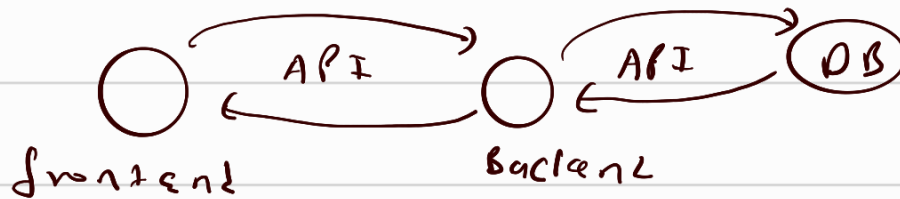


Flask is a framework

* API is set of rules and protocols that allows software to interact with each other



Types of API's

Based on Availability

Public API's: external or developer facing API, available to any 3rd party developer, free (open) or require a subscription (commercial)

Partner API's: shared externally, with specific business partners, usually under contractual agreements

Private API's: private to company only. They will not be shared.

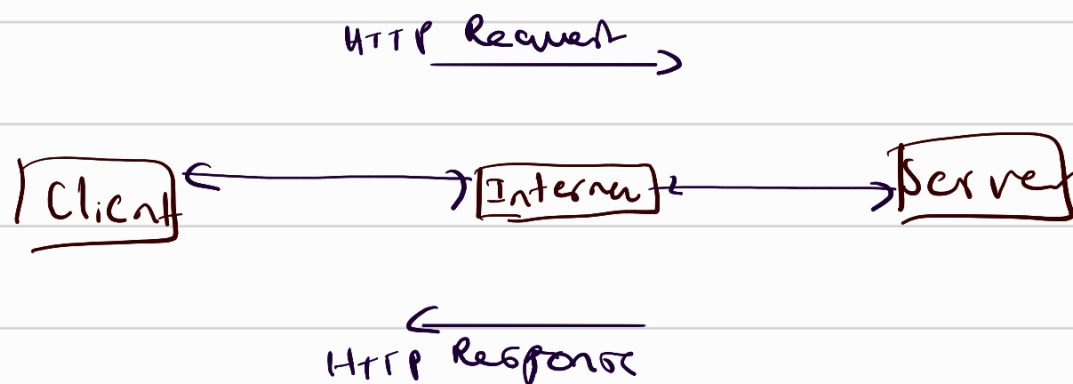
Based on use cases:

Database API: These allow communication b/w an application and a database management system.

web API - exchange of data and functions b/w web based systems, following a client-server architecture.

Web API Concept (16:27)

Client - Server Architecture.



End to End Interaction

API call: Client sends request to server,

Endpoint: Specific URL where the API call is directed

Response = server process the request and sends,

API Protocols = (SOAP, REST, GraphQL)

(a) SOAP (Simple Object Access Protocol)

→ This is for exchanging structured info in web services using XML

→ Highly Standardized, supports only XML format. often used in enterprise applⁿ for secure, transaction-oriented operations.

→ common in payment gateways, CRM solutions and telecom services

(b) REST (Representational State Transfer)

→ This uses standard HTTP methods (GET, POST, PUT, DELETE) for web services.

→ Light weight, supports multiple formats (JSON, XML, Binary etc)

- used in building scalable web services
- ex: social media, online market place

(c) Graph QL

→ a query language for APIs that allow clients to specify what data they need.

→ Precise data, efficient for applications with complex data requirements.

→ Ideal for modern web and mobile apps (46:14)

REST API's

web services that adhere to the REST architecture are called RESTful Services or REST APIs.

→ Client server architecture

→ Stateless

→ Layered system

- ① client
- ② API gateway
- ③ server
- ④ Backend

→ cache ability: Response from the server can be cached to decrease Repeated request to server.

→ Uniform Interface:

HTTP methods

POST: Create a new resource

GET: Retrieve a resource (or) collection

PUT: Updates an existing resource

DELETE: Remove a resource

